# CST MICROWAVE STUDIO TUTORIAL WORKSHOP

**Manual version 1.4**

**Conducted by SUTD Student Chapter (Mu Epsilon)**
**of IEEE–HKN (Eta Kappa Nu)**

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## Introduction

CST Microwave Studio (CST MWS) is a specialist tool for the 3D EM simulation of high frequency components. CST MWS enables the fast and accurate analysis of high frequency (HF) devices such as antennas, filters, couplers, planar and multi-layer structures and SI and EMC effects. Exceptionally user friendly, CST MWS quickly gives you an insight into the EM behavior of your high frequency designs.

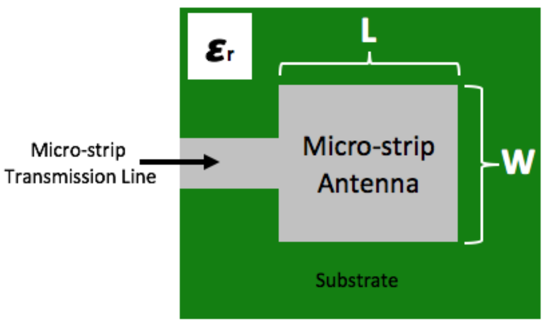
This tutorial aims to educate on using the CST Microwave Studio software to design a WiFi Patch Antenna. There are a total of five parts to this tutorial:

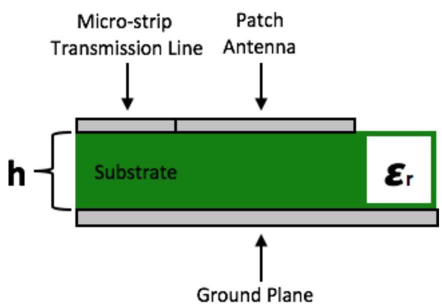
1. Creating the Design project file
2. Modelling of WiFi Patch Antenna
3. Inset Fed Patch Antenna Simulation and Results
4. Parameter Sweeping
5. Generating 2D DXF and 2D Gerber files for PCB Milling Machine

At the end of the tutorial, user should be able to fabricate an actual WiFi Patch Antenna by using the LPKF PCB milling machine in the SUTD Fab Lab with assistance from the CST Microwave Studio software.

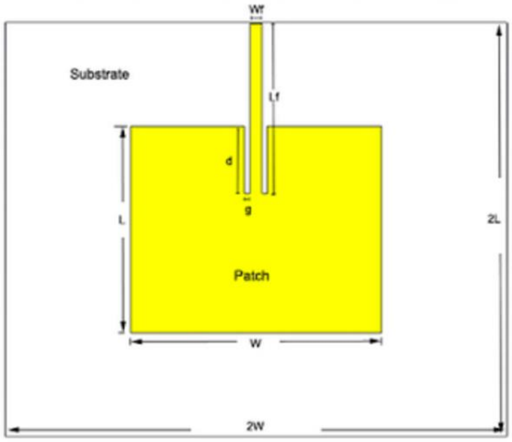
Microstrip or patch antennas are becoming increasingly useful because they can be printed directly onto a circuit board. Microstrip antennas are becoming very widespread within the mobile phone market. Patch antennas are low cost, have a low profile and are easily fabricated. The size of a microstrip antenna is inversely proportional to its frequency. At frequencies lower than microwave, microstrip patches don't make sense because of the sizes required. Microwaves are forms of electromagnetic radiation with wavelengths ranging from one meter to one millimeter; with frequencies between 300 MHz (100 cm) and 300 GHz (0.1 cm).

$$\textit{Size of Microstrip Antenna} \propto \frac{1}{\textit{Frequency}}$$



The patch antenna picture that will be designed is shown below-




***Advantages of microstrip antennas include:***

- Low cost to fabricate.
- Conformal structures are possible (it's easy to form curved surfaces, as long as the curve is in one direction only).
- Easy to form a large array, spaced at half-wavelength or less.
- Light weight.

***Disadvantages include:***

- Limited bandwidth. (A bandwidth is a measurement of the width of a range of frequencies, measured in hertz.)
- Low power handling.

In circuit theory, the impedance is a generalization of electrical resistivity in the case of alternating current, and is measured in ohms. Impedance is the complex-valued generalization of resistance. The impedance of an ideal resistor is purely real and is referred to as a resistive impedance:

$$Z_R = R$$

Ideal inductors and capacitors have a purely imaginary reactive impedance:

The impedance of inductors increases as frequency increases:

$$Z_L = j\omega L$$

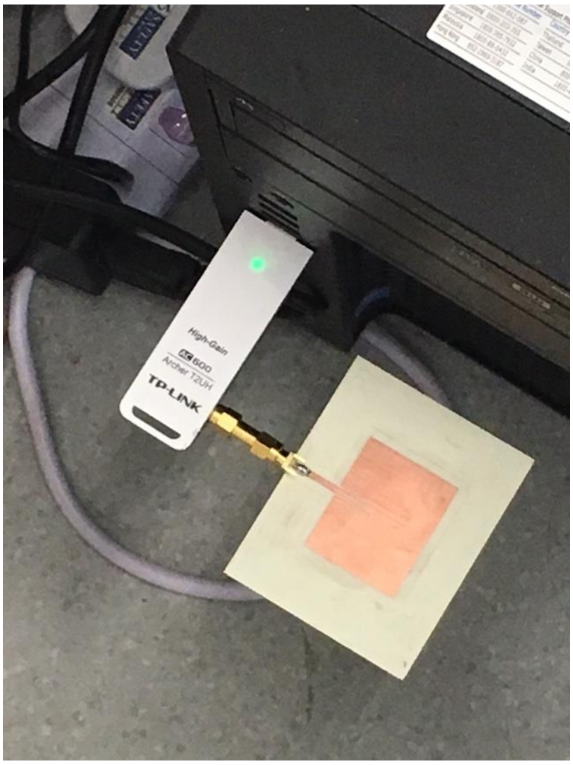
The impedance of capacitors decreases as frequency increases:

$$Z_C = \frac{1}{j\omega C}$$

Impedance matching is important when components of an electric circuit are connected (waveguide to antenna for example): The impedance ratio determines how much of the wave is transmitted forward and how much it is reflected. The reflection coefficient can be calculated using:

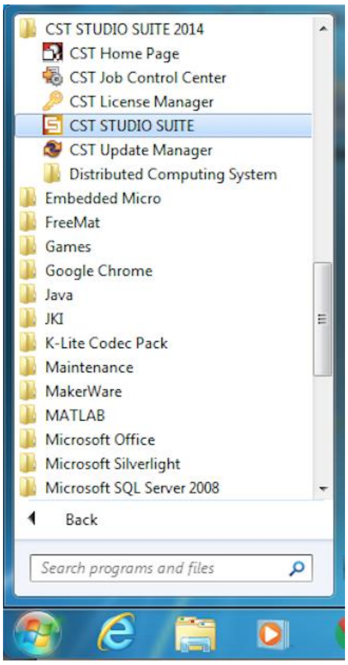
$$\Gamma = \frac{Z_2/Z_1 - 1}{Z_2/Z_1 + 1}$$

, where $\Gamma$ is the reflection coefficient (0 denotes full transmission, 1 full reflection, and 0.5 is a reflection of half the incoming voltage), $Z_1$ and $Z_2$ are the impedance of the first component (from which the wave enters) and the second component, respectively. An impedance mismatch creates a reflected wave.

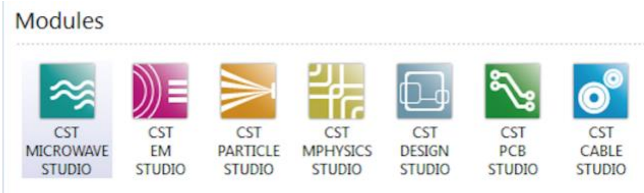


## Part I: Creating the Design project file

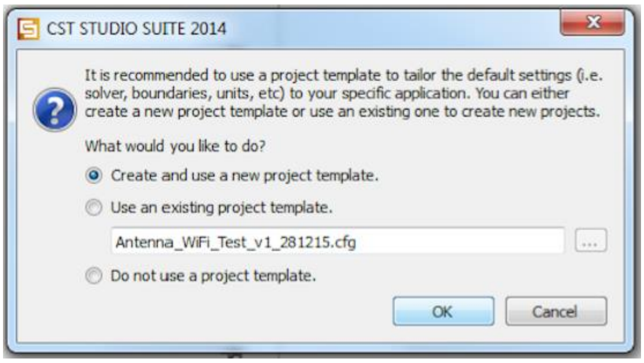
**Step 1:** Launch the "CST STUDIO SUITE" program from the Windows 8 Start Menu under the "CST STUDIO SUITE 2014" folder.



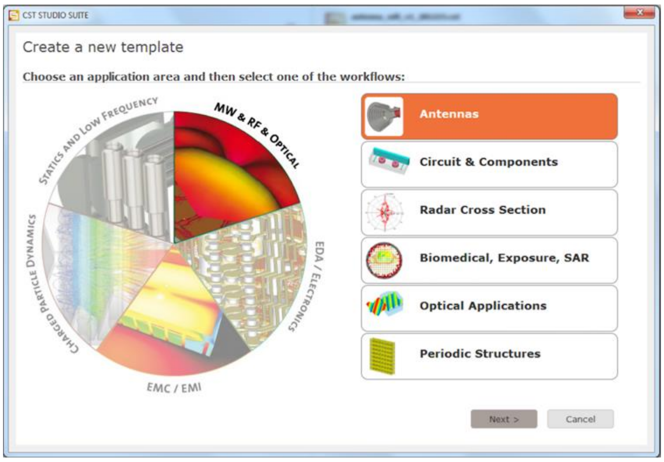

**Step 2:** Select "CST MICROWAVE STUDIO" under the *Modules* section.



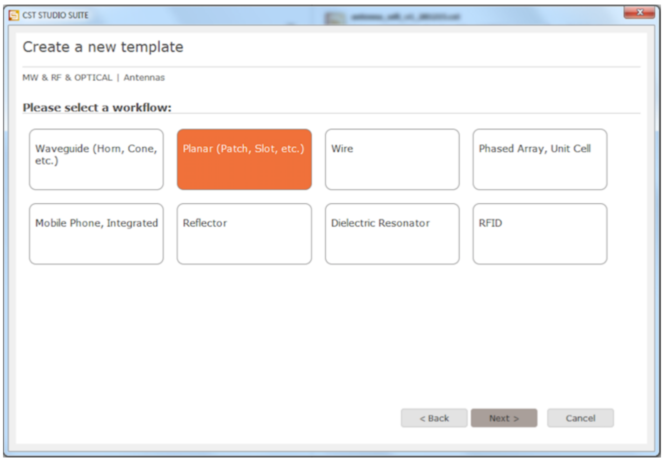

**Step 3:** Check the *radio button* for "Create and use a new project template" and click "OK" to create a new project file.



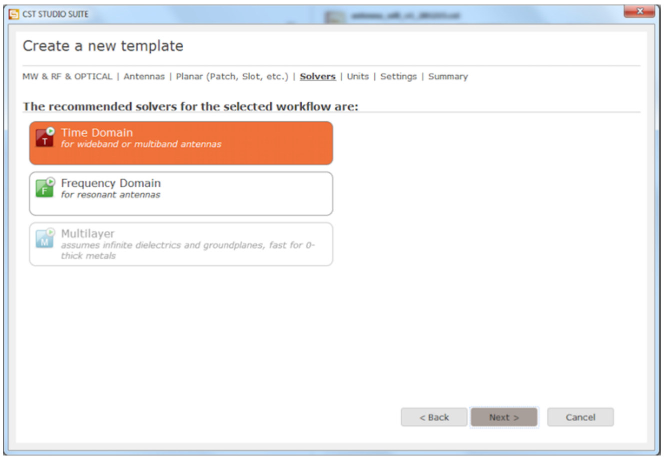

**Step 4:** Select "MW & RF Optical" from the pie chart and then select "Antennas" for designing of 2.45Ghz WiFi Antenna which operates in the Microwave Band. Click "Next" to proceed.



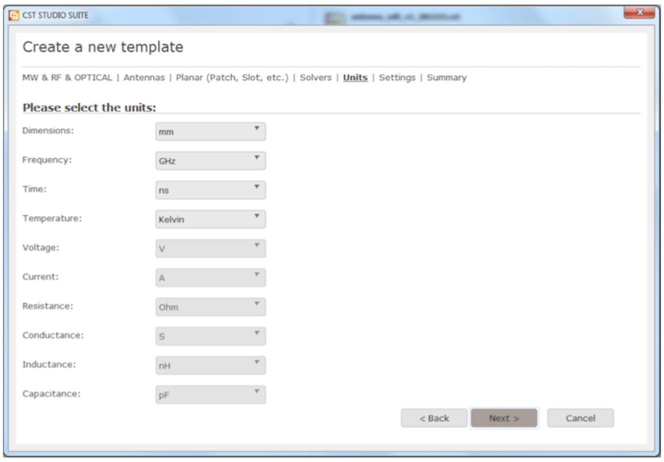

**Step 5:** Select "Planar (Patch, Slot, etc.)" for designing of WiFi Patch Antenna. Click "Next" to proceed.



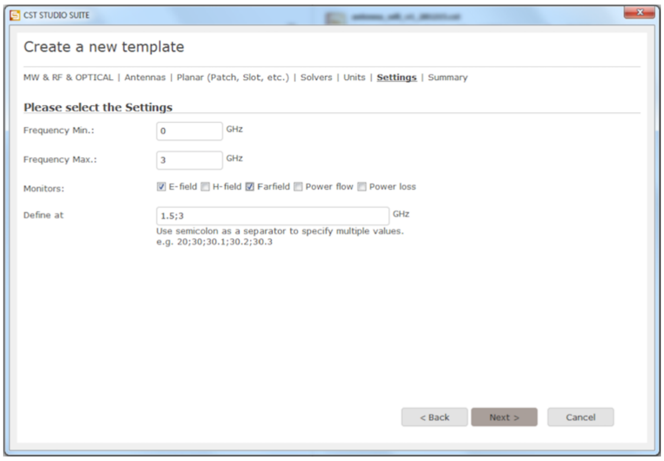
**Step 6:** Select "Time Domain". Click "Next" to proceed.



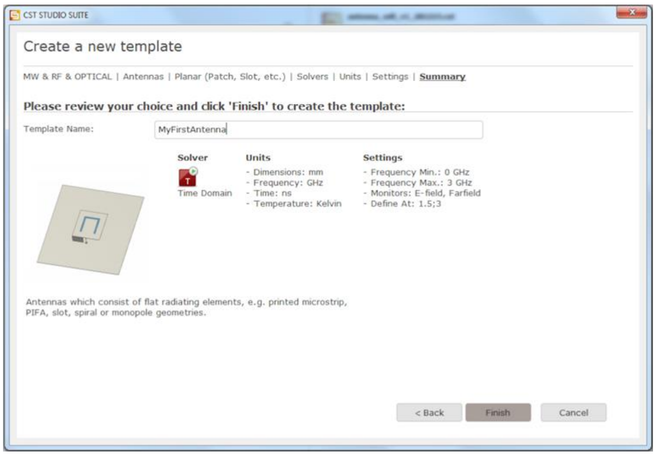

**Step 7:** Change your *Dimensions* to the "mm" scale and *Frequency* to the "GHz" scale for greater software user-friendliness. Click "Next" to proceed.




**Step 8:** Set *Frequency Min.* to "0GHz" and *Frequency Max.* to "3GHz". This band of frequencies will be inspected during the simulation analysis later on with "2.45GHz" falling within this band-of-interest. Also, check on the "E-field" and "Farfield" boxes. Click "Next" to proceed.




**Step 9:** Name your project. Click "Finish" to create your design project file.

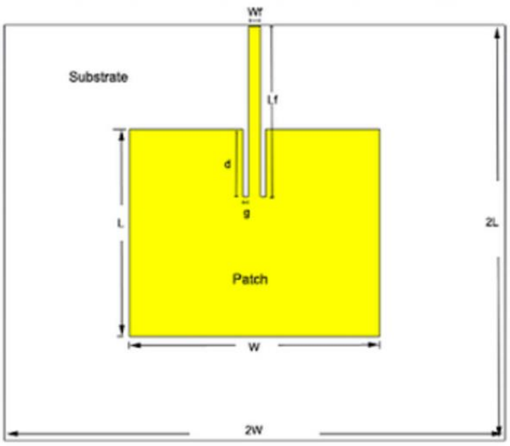



**< End of Part I >**

## Part II: Modelling of WiFi Patch Antenna

**Step 1:** W and L are the width and length of the WiFi Patch Antenna. The substrate has twice the width and length of the patch antenna, that is, 2W and 2L. Wf and Lf are the dimensions of the microstrip line. “d” is the inset fed distance and “g” is the gap between the patch antenna and the microstrip feed line.

The patch antenna picture that will be designed is shown below-




**Step 2:** Go on to [https://www.pasternack.com/t-calculator-microstrip-ant.aspx](https://www.pasternack.com/t-calculator-microstrip-ant.aspx) to auto-generate the width (W) and length (L) of the WiFi Patch Antenna according to the parameters of our purchased PCB laminate.

| **RO4350 Laminate** |
| --- |
| Dielectric Constant: 3.48 |
| Dielectric Height (Thickness): 0.762mm |
| Operation Frequency: 2.45GHz |

Next, under *Parameter List* (near the bottom left-hand corner of the screen), define the following variables of the WiFi Patch Antenna. Theories and formula related to the following parameters will be covered in greater details during undergraduate level engineering programme in SUTD (course 30.102: Electromagnetics and Applications). <u>Do not forget to fill in the red boxes below with your auto generated L and W values.</u>

Parameter List

| Name | Value | Description | Type |
| --- | --- | --- | --- |
| d | 12.7 | Microstrip Feed | Length |
| g | 0.5 | Microstrip Feed | Length |
| h | 0.762 | Substrate height | Length |
| L | | Patch Antenna | Length |
| Lf | 32 | Microstrip Feed | Length |
| t | 0.07 | Patch Antenna thic | Length |
| W | | Patch Antenna | Length |
| Wf | 2.3 | Microstrip Feed | Length |
| | | | Undefined |

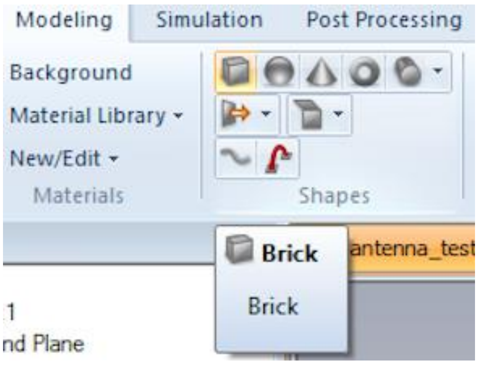
## Creating the Substrate

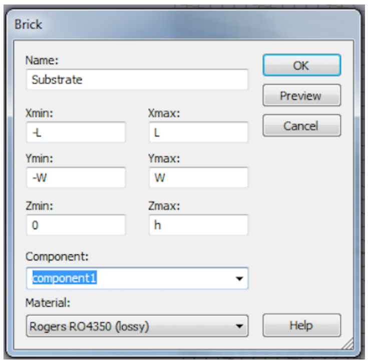
**Step 3:** From the *Modelling* section of the CST program, select the “Brick” icon and press “ESC” on your keyboard. This will bring up a window that allows users to specify the dimensions of the substrate.

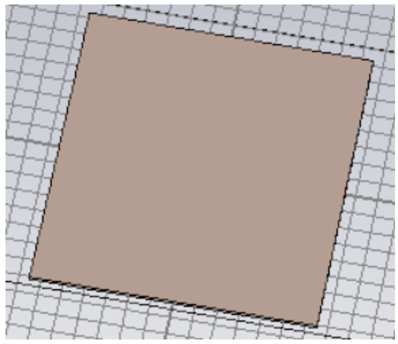
Give a name “**Substrate**” and enter the following:

**Length (Xmin = -L, Xmax = L)**
**Width (Ymin = -W, Ymax = W)**
**Height (Zmin = 0, Zmax = h)**

From the dropdown menu of the *Material* section, select “Load from Material Library” and search for Rogers RO4350 (lossy). Press “Load” and then “OK” to complete the substrate definition. Now, the substrate will appear on the screen.

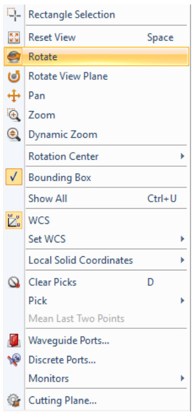




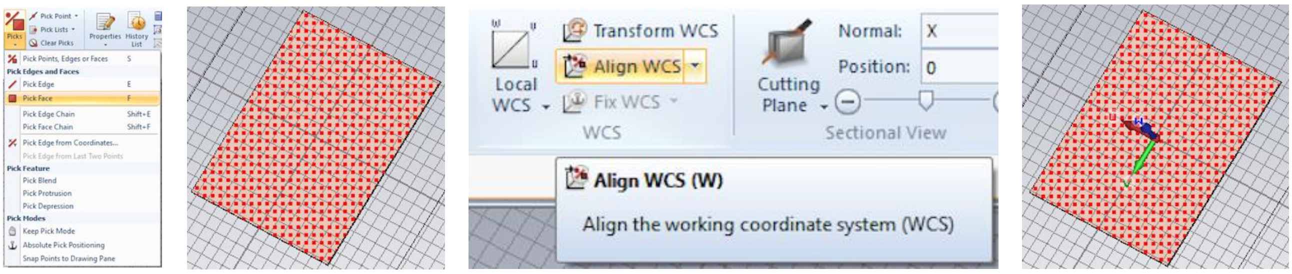


## Creating the Ground Plane

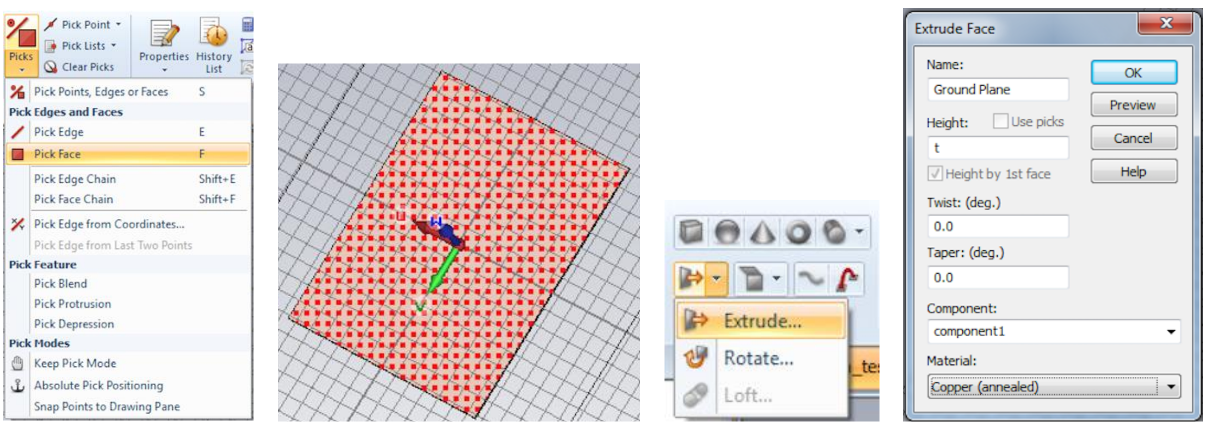
**Step 4:** Rotate the substrate to show the back face of it. To enter the “Rotate” mode, right click on the screen and select “Rotate”.




From the *Modelling* section of the CST program, under “Picks”, select “Pick Face”. Double click on the back face of the substrate to *Pick* this back face. Once the back face is highlighted, from the *Modelling* section of the CST program, click “Align WCS” to shift the drawing/modelling reference on to the back face of the substrate.

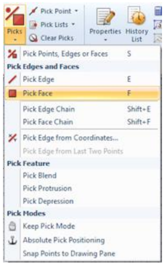
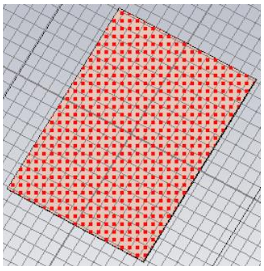
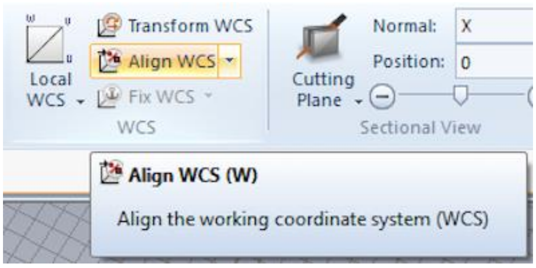


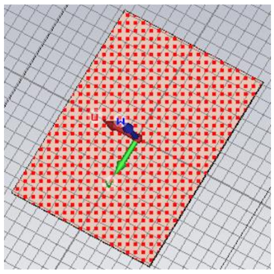

**Step 5:** Again, from the *Modelling* section of the CST program, under “Picks”, select “Pick Face” and pick the back face of the substrate by double clicking on this back face of the substrate. Now, select the “Extrude” icon under the “Brick” icon. Name the new surface that will be extruded as “Ground Plane” then enter the Height as “ **t** ” and from the dropdown menu of the *Material* section, select “Load from Material Library” and search for Copper (annealed). Press “Load” and then “OK” to complete the ground plane definition. Now, the ground plane will be extruded from the substrate.



## Creating the Patch Antenna

**Step 6:** Rotate the substrate to show the top face of it. From the *Modelling* section of the CST program, under "Picks", select "Pick Face". Pick the top face of the substrate. Once the top face is highlighted, from the *Modelling* section of the CST program, click "Align WCS" to shift the drawing/modelling reference on to the top face of the substrate.

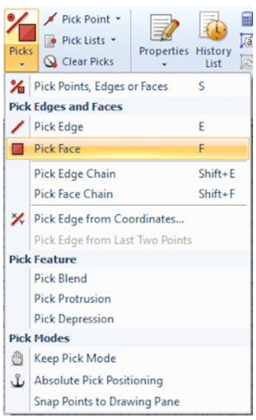


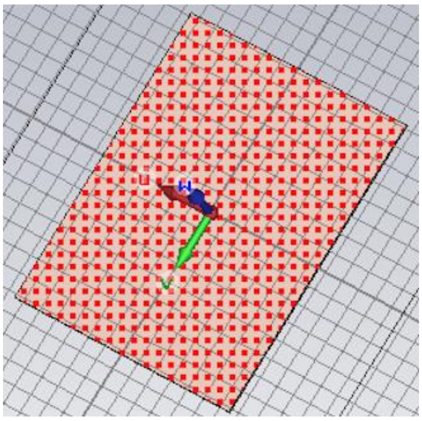
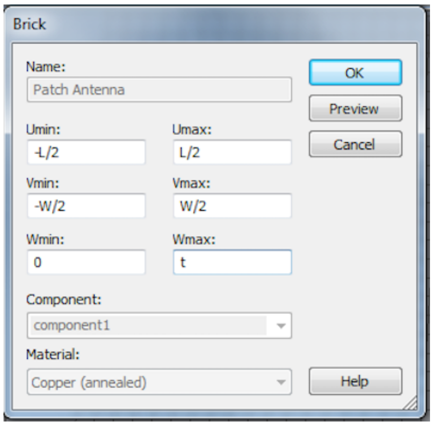
**Step 7:** From the *Modelling* section of the CST program, select the "Brick" icon and press "ESC" on your keyboard.

Give a name "**Patch Antenna**" and enter the following:

**Length (Umin = -L/2, Umax = L/2)**
**Width (Vmin = -W/2, Vmax = W/2)**
**Height (Wmin = 0, Wmax = t)**

From the dropdown menu of the *Material* section, select "Load from Material Library" and search for Copper (annealed). Press "Load" and then "OK" to complete the patch antenna definition. Now, the patch antenna will be built on the top face of the substrate.



## Creating the Microstrip Feed Line

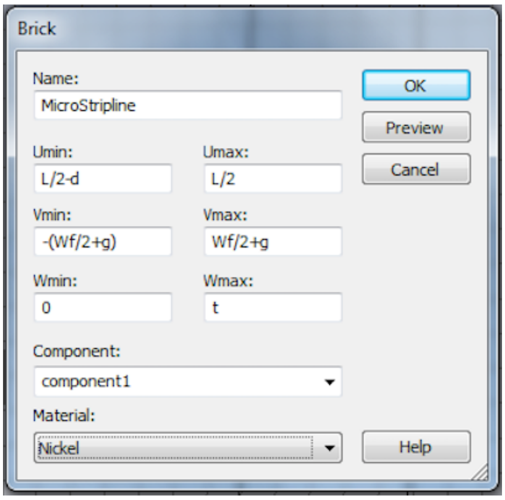
**Step 8:** To create a microstrip feed line, first an empty space will need to be created before adding in the microstrip feed line. This creates a gap between the microstrip feed line and the patch antenna. From the *Modelling* section of the CST program, select the "Brick" icon and press "ESC" on your keyboard.

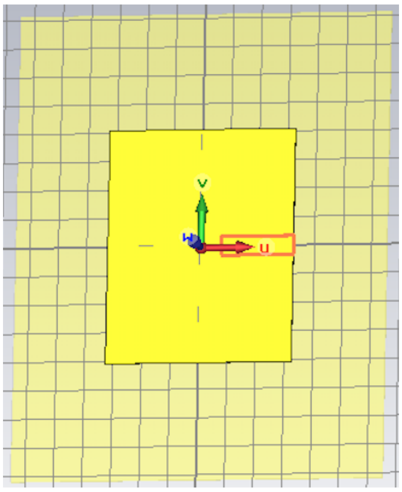
Give a name "**substractor**" and enter the following:

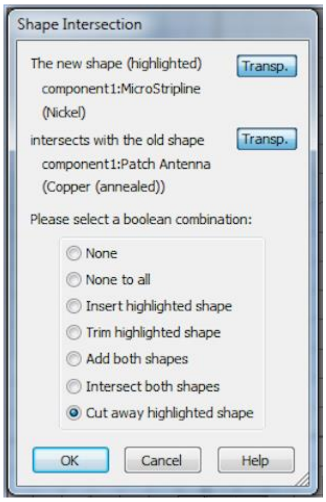
**Length (Umin = L/2-d, Umax = L/2)**
**Width (Vmin = -(Wf/2+g), Vmax = Wf/2+g)**
**Height (Wmin = 0, Wmax = t)**

From the dropdown menu of the *Material* section, select "Load from Material Library" and search for Nickel (or any default material. It doesn't matter which material you are using). Press "Load" and then "OK" to complete the substractor definition.

Once the above is done, a *Shape Intersection* window will appear because the newly created "substractor" shape has intersected with the "Patch Antenna" shape. Check the "Cut away highlighted shape" *radio button* and create the empty space.



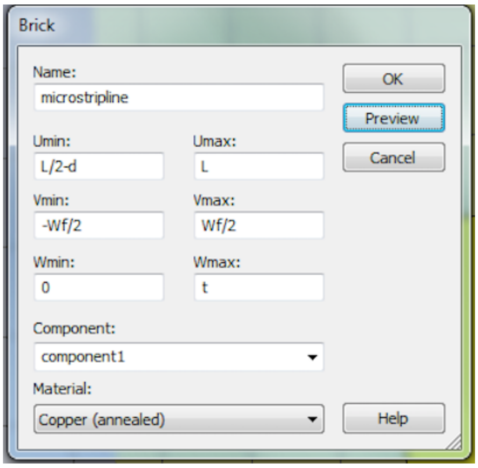




**Step 9:** From the *Modelling* section of the CST program, select the "Brick" icon and press "ESC" on your keyboard.

Give a name "**microstripline**" and enter the following:

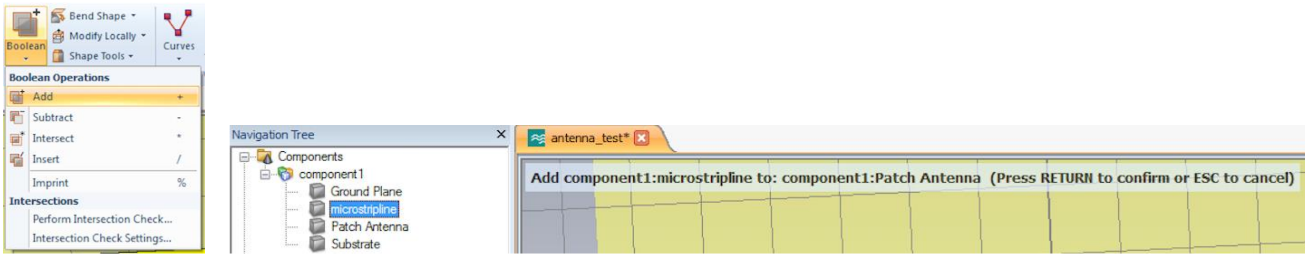
**Length (Umin = L/2-d, Umax = L)**
**Width (Vmin = -Wf/2, Vmax = Wf/2)**
**Height (Wmin = 0, Wmax = t)**

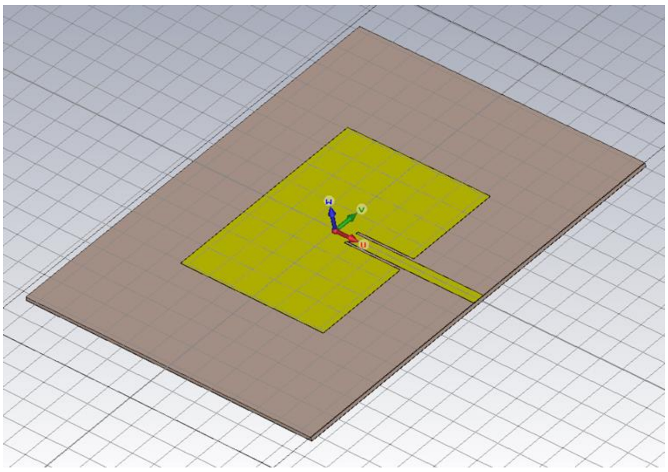
From the dropdown menu of the *Material* section, select Copper (annealed). Press "OK" to complete the microstrip feed line definition. Now, the microstrip feed line will be built on the top face of the substrate.




## Merging the Patch Antenna with the Microstrip Feed Line

**Step 10:** Under *Navigation Tree* (on the left-hand side of the screen), expand "Components" >>> "component1" and single click on "Patch Antenna". From the *Modelling* section of the CST program, under "Boolean", select "Add" and then, single click on " **microstripline** ". Hit "Enter" on your keyboard to merge the patch antenna shape and microstrip feed line shape into a single shape. The final design is shown below.

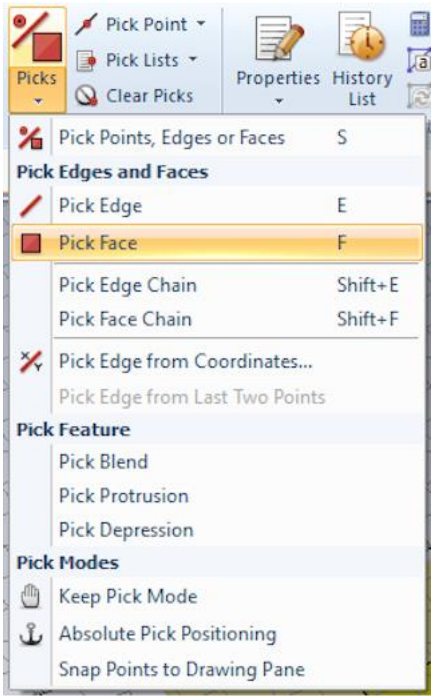


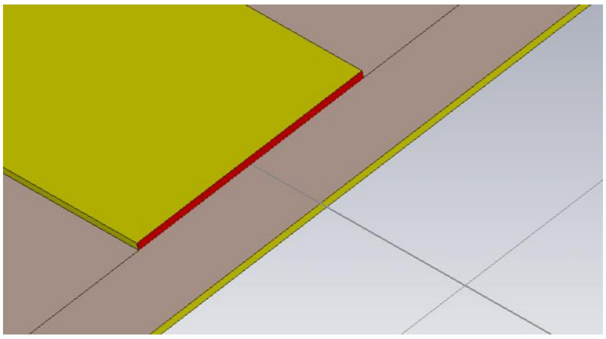



**< End of Part II >**

## Part III: Inset Fed Patch Antenna Simulation and Results

**Step 1:** From the *Modelling* section of the CST program, under "Picks", select "Pick Face". Pick the face shown in red below. This is the face where the waveguide port will operate. In this case, it is the face of the microstrip feed line where the signal will enter.



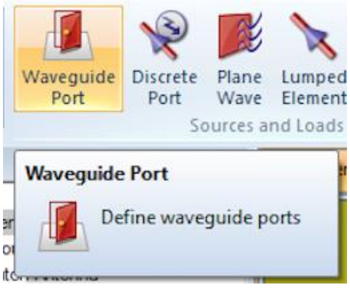



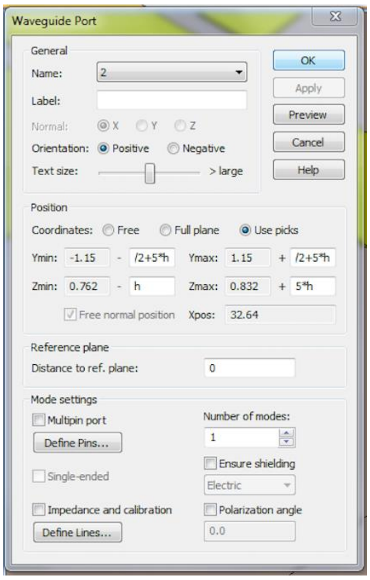
**Step 2:** From the *Simulation* section of the CST program, select "Waveguide Port". A window will be opened for setting up of port dimensions.

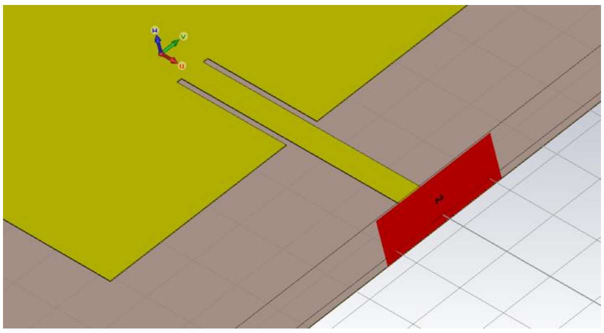
Set:

**Ymin = Wf/2+5*h, Ymax = Wf/2+5*h,**
**Zmin = h, Zmax = 5*h**

Leave other settings as default as shown in the figure below. Click "OK" to achieve the port in red below.



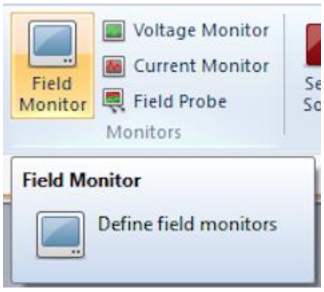



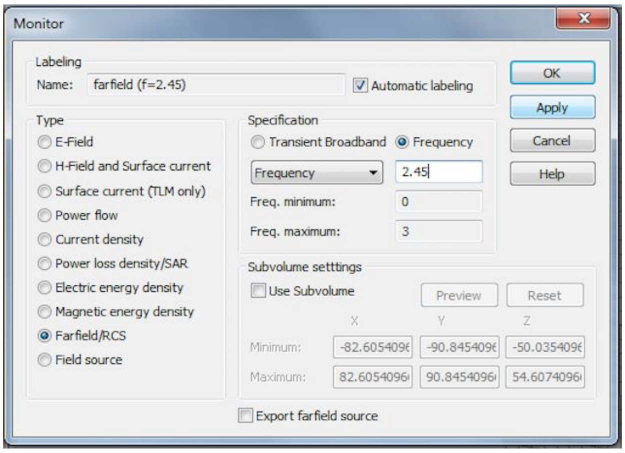



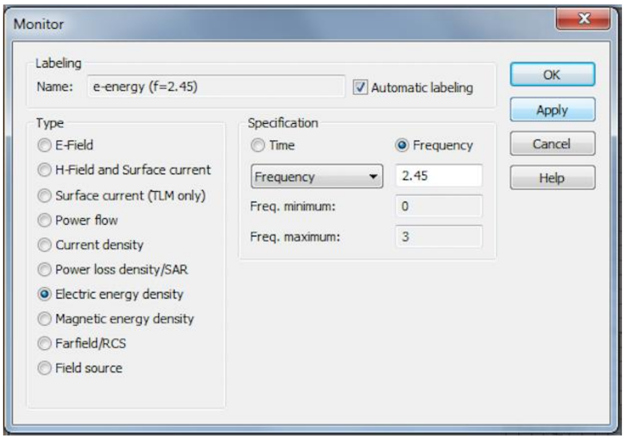
**Step 3:** For this workshop, we are (i) measuring the *S11 or Return Loss* and (ii) viewing the *far field radiation* from the 2.45Ghz WiFi Patch Antenna.

From the *Simulation* section of the CST program, click on "Field Monitor". Then from the field monitor window, check the "Farfield/RCS" *radio button* and then set the *frequency* as 2.45Ghz. Click "Apply".

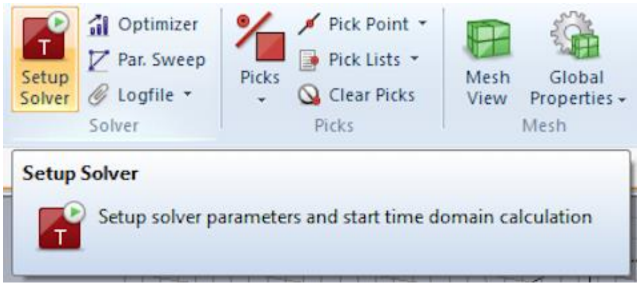
Next, check the "Electric energy density" *radio button* and then set the *frequency* as 2.45Ghz. Click "OK" to end the setup.

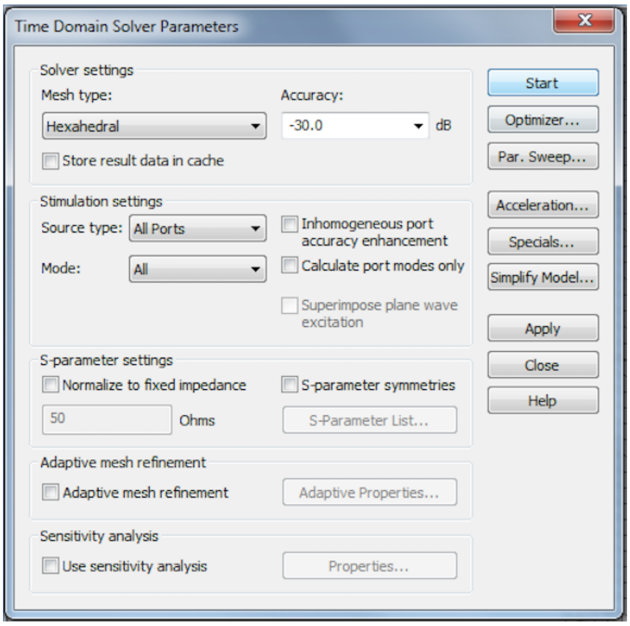


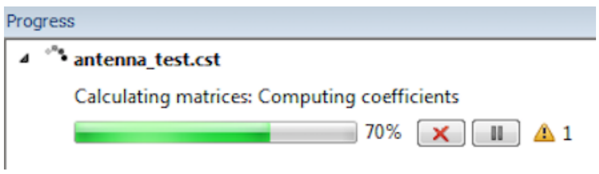




**Step 4:** From the *Simulation* section of the CST program, click on "Setup Solver". This brings up the dialog box. While leaving all settings as default, click "Start" to start time domain calculation. This may take some time.

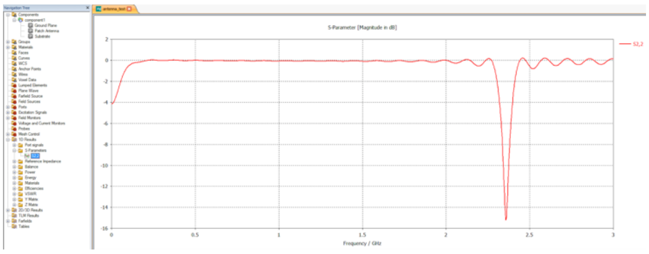


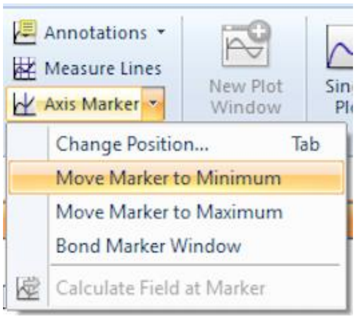



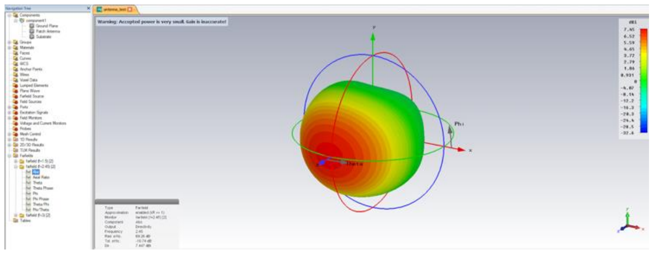
**Step 5:** Under *Navigation Tree*, select “1D Results” >>> “S-Parameters” >>> “S1,1”.

From the *1D Plot* section of the CST program, select “Axis Marker” >>> “Move Marker to Minimum” to check the minimum value in the plot. For a good *Return loss*, the value should fall below -16dB to -20dB. The lower the return loss, the better the antenna.






**Step 6:** Under *Navigation Tree*, select “Farfields” >>> “farfield (f=2.45)”.

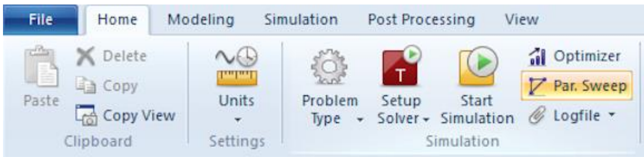


**< End of Part III >**

## Part IV: Parameter Sweeping

One of the techniques which allows us to improve on the simulated performance of the WiFi Antenna is to sweep a range of the controlled parameter(s) numerically (by brute force) to know the performance due to different values of the parameter(s) within the defined range. (Another technique is to use the "Optimizer" function from the *Home* section of the CST program.) We will now use the "Parameter Sweep" function.

**Step 1:** From the *Home* section of the CST program, select "Par. Sweep".

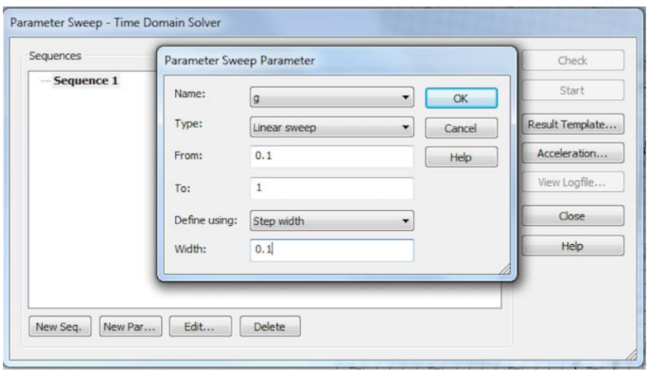



**Step 2:** Click "New Seq." followed by "New Par…".

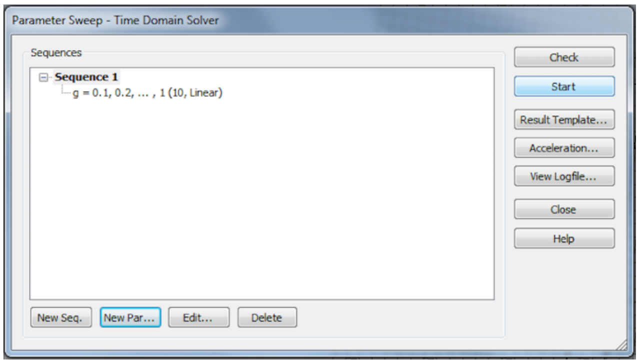
Select "*Name:*" of the parameter to be "g" using the dropdown menu. We will tune the performance with "g".

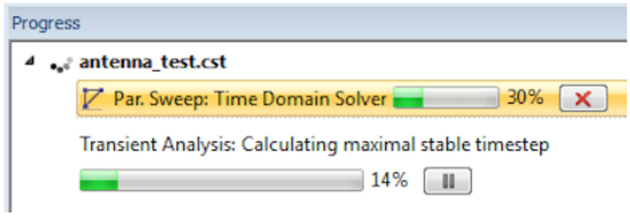
Select "*Type:*" to be "Linear sweep" using the dropdown menu.

Select "*Define using:*" to be "Step width" using the dropdown menu. For our preference, we will sweep from "*0.1mm*" to "*1mm*" with a Width step size of "*0.1mm*". Note: A smaller step size will result in a longer solving time as the solving resolution is increased.




**Step 3:** Press "Start" to start evaluating the performance numerically according to different values of the parameters within the defined range set by you. This may take some time.

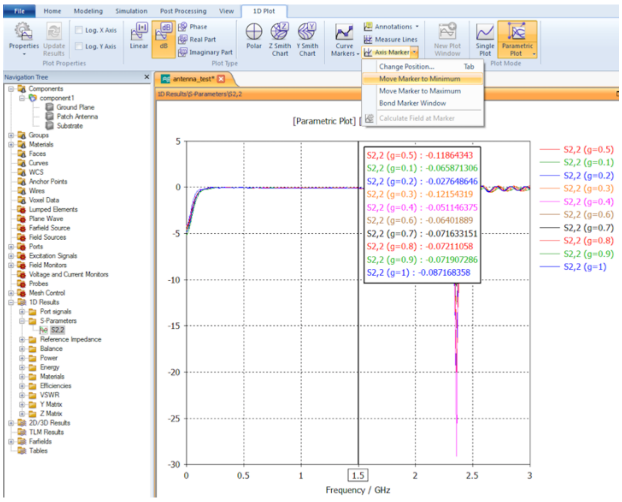



**Step 4:** Under *Navigation Tree*, select "1D Results" >>> "S-Parameters" >>> "S1,1".

From the *1D Plot* section of the CST program, select "Parametric Plot" to view the S11 performance due to the range of values of "g" all under one plot. From the plot, it can be concluded that g = 0.4mm provides the best performance. You may change the value of g from 1mm to 0.4mm from the *Parameter List* table.

Side note: You may try out the parameter sweep with other parameters in order to get better general performance.

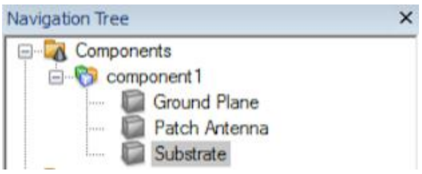



**< End of Part IV >**

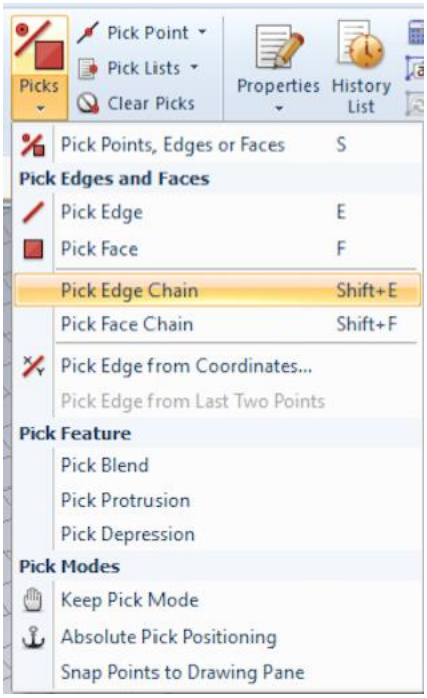
## Part V: Generating 2D DXF and 2D Gerber files for PCB Milling Machine

We would like to generate the 2D **DXF** file to allow the PCB Milling Machine to cut the PCB laminate according to the perimeter of the antenna.

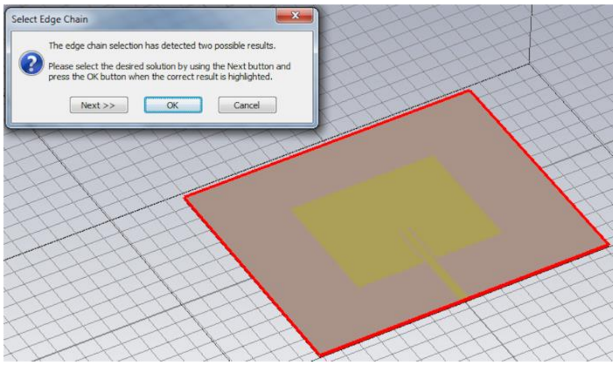
**Step 1:** Under *Navigation Tree*, select "Substrate".



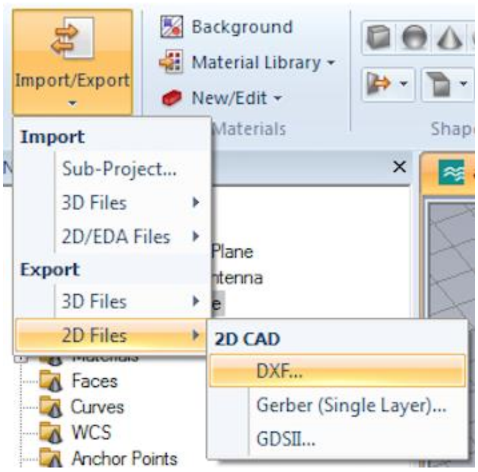

**Step 2a:** From the *Modelling* section of the CST program, under "Picks", select "Pick Edge Chain". (You may select "Pick Edge" instead to select the 4 edges of the substrate material one-by-one, which is not recommended.)




**Step 2b:** Double click on the edge of the substrate material. The perimeter should be highlighted in red. Click "OK" to proceed.

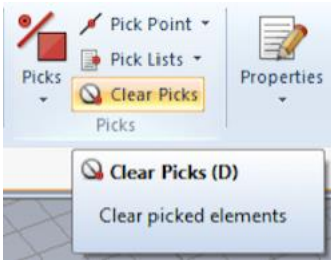



**Step 3:** From the *Modelling* section of the CST program, under "Import/Export", select "2D Files" >>> "DXF…" to save the 2D DXF file of the substrate material to your desired location on the computer.

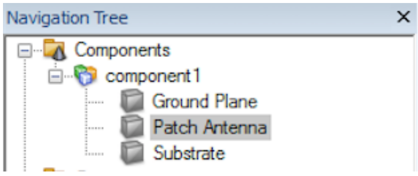


Lastly, we would like to generate the 2D **Gerber** file to allow the PCB Milling Machine to mill the necessary areas beside the actual patch antenna. The Gerber format is an open 2D binary vector image file format. It is the de facto standard used by printed circuit board (PCB) industry software to describe the printed circuit board images: copper layers, solder mask, legend, etc.

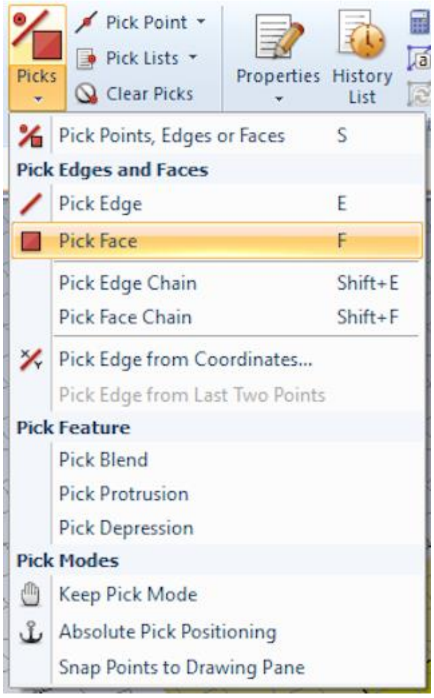
**Step 4:** From the *Modelling* section of the CST program, select "Clear Picks" to unpick the previous edges.



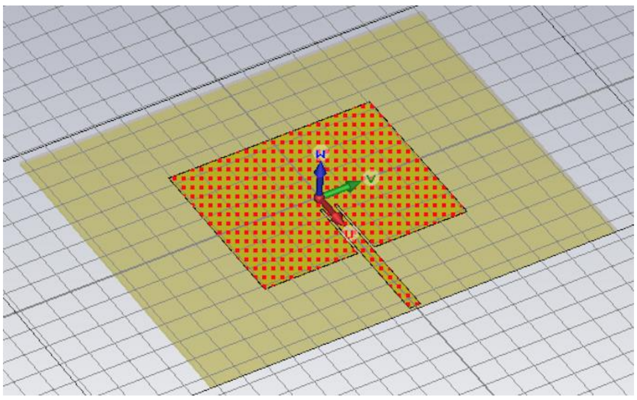

**Step 5:** Under *Navigation Tree*, select "Patch Antenna".



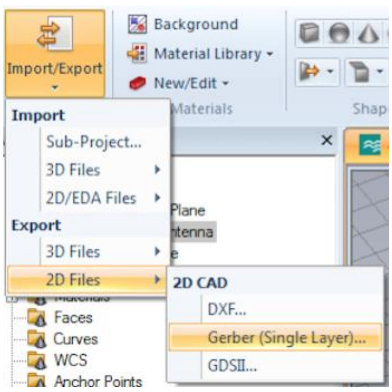

**Step 6a:** From the *Modelling* section of the CST program, under "Picks", select "Pick Face".




**Step 6b:** Double click on the face of the patch antenna. The patch antenna area should be highlighted in red.



**Step 7:** From the *Modelling* section of the CST program, under "Import/Export", select "2D Files" >>> "Gerber (Single Layer)…" to save the 2D Gerber (.gbr) file of the patch antenna to your desired location on the computer.



**< End of Part V >**

## Endnote

Users of CST Microwave Studio are given great flexibility in tackling a wide application range through the variety of available solver technologies. CST Microwave Studio is seen by an increasing number of engineers as an industry standard development tool.

For more details on how to utilize CST Microwave Studio, you may refer to the tutorial on http://eee.guc.edu.eg/Courses/Communications/COMM905%20Advanced%20Communication%20Lab/Sessions/MWS_Tutorials.pdf.

## Reference

Bevelacqua, P. (n.d.). Antenna-Theory. Retrieved December 1, 2016, from http://www.antenna-theory.com/antennas/patches/antenna.php